

# Multi-dimensional equilibria preserving methods for hyperbolic equations

**Davide Torlo**, Wasilij Barsukow, Mirco Ciallella, Mario Ricchiuto, Moussa Ziggaf

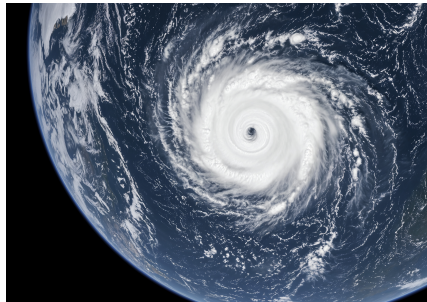
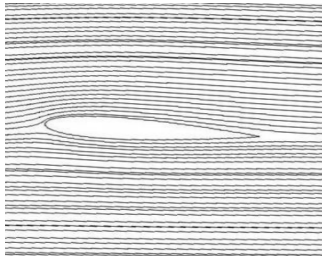
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# Moving equilibria in hyperbolic problems



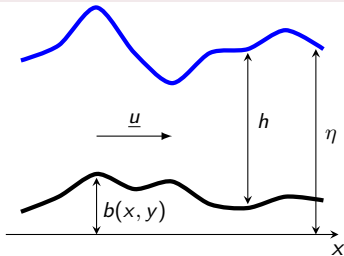
# Moving equilibria in hyperbolic problems

## Hyperbolic Balance Laws in 2D

$$\partial_t q + \partial_x f(q) + \partial_y g(q) = S(q, x, y, t)$$

## Equilibria for Balance Laws in 2D

$$\partial_x f(q) + \partial_y g(q) = S(q, x, y, t)$$



## Shallow Water Equations

$$\begin{cases} \partial_t h + \partial_x(hu) + \partial_y(hv) = 0 \\ \partial_t(hu) + \partial_x(hu^2 + \frac{g}{2}h^2) + \partial_y(huv) = -gh\partial_x b \\ \partial_t(hv) + \partial_x(huv) + \partial_y(hv^2 + \frac{g}{2}h^2) = -gh\partial_y b \end{cases}$$

## SW equilibria

- Lake at rest:  $u = v \equiv 0$ ,  $h + b \equiv \eta_0$  constant
- 1D moving equilibria:  $hu \equiv q_0$  constant,  $h$  such that  $\frac{q_0^2}{2gh^2} + h + b \equiv \eta_0$  constant
- 2D moving equilibria:  $\partial_x(hu) + \partial_y(hv) = 0$  and balancing of  $h$ , e.g. vortices

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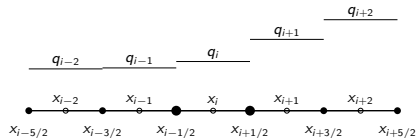
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- 3 Finite Volume Global Flux  
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## Global Flux in 1D for FV 1st order<sup>123</sup>

$$\partial_t q + \partial_x f(q) = S(q, x) \quad \implies \quad \partial_t q + \partial_x G(q, x) = 0$$

$$\text{with } G(q, x) := f(q) - K(q, x) = f(q) - \int_{x_0}^x S(q(s), s) ds.$$

$$\text{FV: } q_i \approx \int_{x_{i-1/2}}^{x_{i+1/2}} q(x) dx$$



<sup>1</sup>Gascón, L., Corberán, J. J. Comput. Phys. 172(1), 261–297 (2001)

<sup>2</sup>Chertock, A., Kurganov, A., Liu, X., Liu, Y., & Wu, T. (2022). Journal of Scientific Computing, 90, 1-21.

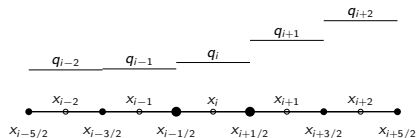
<sup>3</sup>Ciallella, M., Torlo, D., & Ricchiuto, M. (2023). Journal of Scientific Computing, 96(2), 53.

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$$f_i := \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} f(q(x, t)) dx,$$

$$K_i \approx K(x_i, q(x_i)) = \int_{x_0}^{x_i} S(q(s), s) ds \approx K_{i-1} + \int_{x_{i-1}}^{x_i} S(q(s), s) ds,$$

$$K_i := K_{i-1} + \Delta x \frac{S_{i-1} + S_i}{2}$$

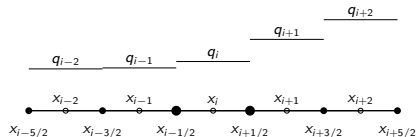
$$G_i := f_i - K_i.$$

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Numerical flux depends only on  $G$ :  
upwind, Roe, NO Rusanov

$$\partial_t q_i + \frac{\hat{G}_{i+1/2} - \hat{G}_{i-1/2}}{\Delta x} = 0,$$

$$\hat{G}_{i+1/2} := \text{sign}(J)^+ G_i + \text{sign}(J)^- G_{i+1},$$

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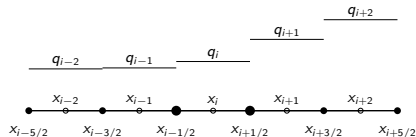
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$$\hat{G}_{i+1/2} := \text{sign}(J)^+ G_i + \text{sign}(J)^- G_{i+1},$$

Equilibrium:  $\hat{G}_{i+1/2} = \hat{G}_{i-1/2} = \hat{G}_0$  for all  $i$

$$f_i - K_i = G_0$$

Mind: high order, other equilibria (LAR), super convergence

$$f_i := \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} f(q(x, t)) dx,$$

$$K_i \approx K(x_i, q(x_i)) = \int_{x_0}^{x_i} S(q(s), s) ds \approx K_{i-1} + \int_{x_{i-1/2}}^{x_i} S(q(s), s) ds,$$

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## Validation: Subcritical flow and perturbation

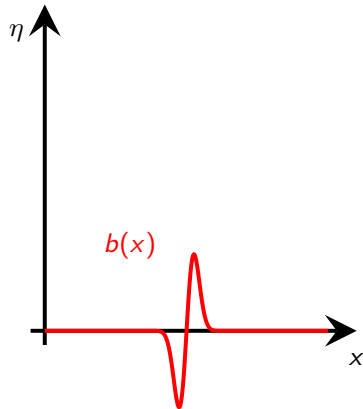
### Domain and Bathymetry

$$\begin{aligned}\Omega &= [0, 25], \\ b(x) &= 0.05 \sin(x - 12.5) \exp(1 - (x - 12.5)^2), \\ g &= 9.812.\end{aligned}$$

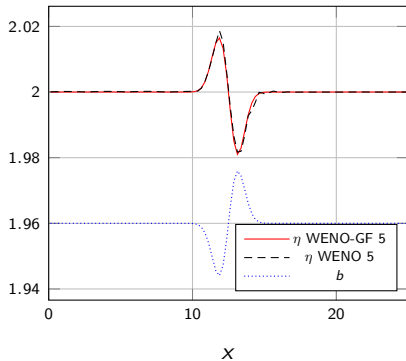
$b(x)$  is chosen  $C^\infty$  and such that it has values smaller than machine precision at the boundaries.

### Subcritical flow test

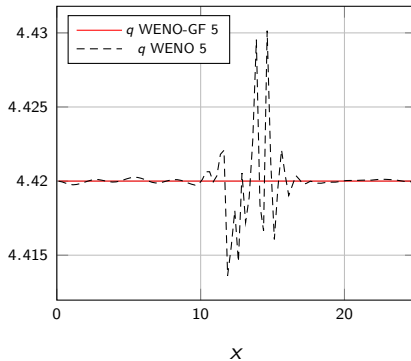
$$\begin{aligned}\text{IC: } h(x, 0) &= 2 - b(x), & q(x, 0) &\equiv 0, \\ \text{BC: } h(25, t) &= 2, & q(0, t) &= 4.42,\end{aligned}$$



## Validation: Subcritical flow and perturbation



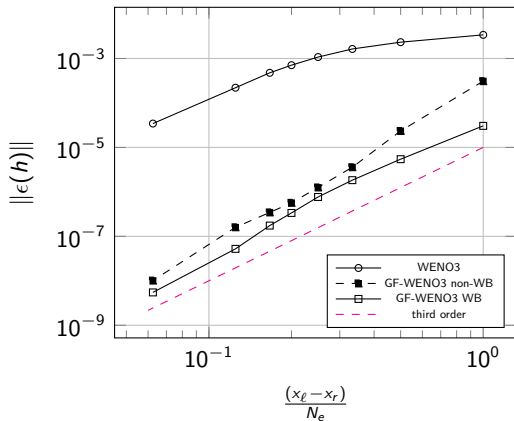
(a)  $\eta$



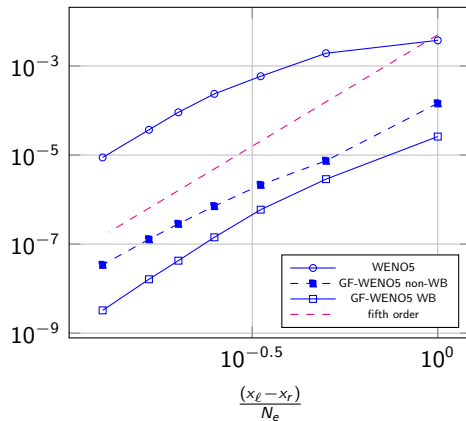
(b)  $q$

**Figure:** Subcritical flow: characteristic variables compute by means of the GF-WENO5 (red continuous line) and WENO5 (black dashed line) schemes with  $N_e = 100$ .

## Validation: Subcritical flow and perturbation



(a) WENO3



(b) WENO5

Figure: Subcritical flow: convergence tests with WENO3 and WENO5.

## Validation: Subcritical flow and perturbation

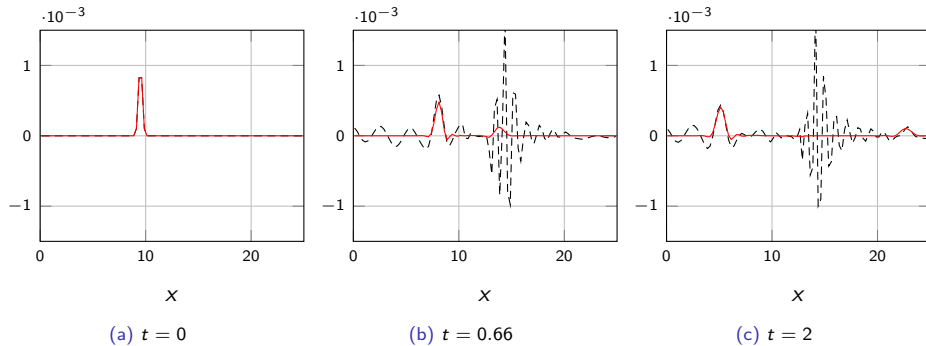


Figure: Perturbation on a subcritical flow:  $\eta - \eta^{eq}$

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### Acoustics

$$\partial_t u + \partial_x p = S_u$$

$$\partial_t v + \partial_y p = S_v$$

$$\partial_t p + \partial_x u + \partial_y v = S_p$$

## SUPG FEM for acoustics

$$\int \varphi(\partial_t u + \partial_x p + S_u) + \alpha \Delta \partial_x \varphi(\partial_t p + \partial_x u + \partial_y v + S_p) = 0$$

$$\int \varphi(\partial_t v + \partial_y p + S_v) + \alpha \Delta \partial_y \varphi(\partial_t p + \partial_x u + \partial_y v + S_p) = 0$$

$$\begin{aligned} \int \varphi(\partial_t p + \partial_x u + \partial_y v + S_p) &+ \alpha \Delta \partial_x \varphi(\partial_t u + \partial_x p + S_u) \\ &+ \alpha \Delta \partial_y \varphi(\partial_t v + \partial_y p + S_v) = 0 \end{aligned}$$

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$$\int \varphi(\partial_t \mathbf{v} + \partial_y \mathbf{p} + S_v) + \alpha \Delta \partial_y \varphi(\partial_t \mathbf{p} + \partial_x \mathbf{u} + \partial_y \mathbf{v} + S_p) = 0$$

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### Acoustics

$$\partial_t \mathbf{u} + \partial_x \mathbf{p} = S_u$$

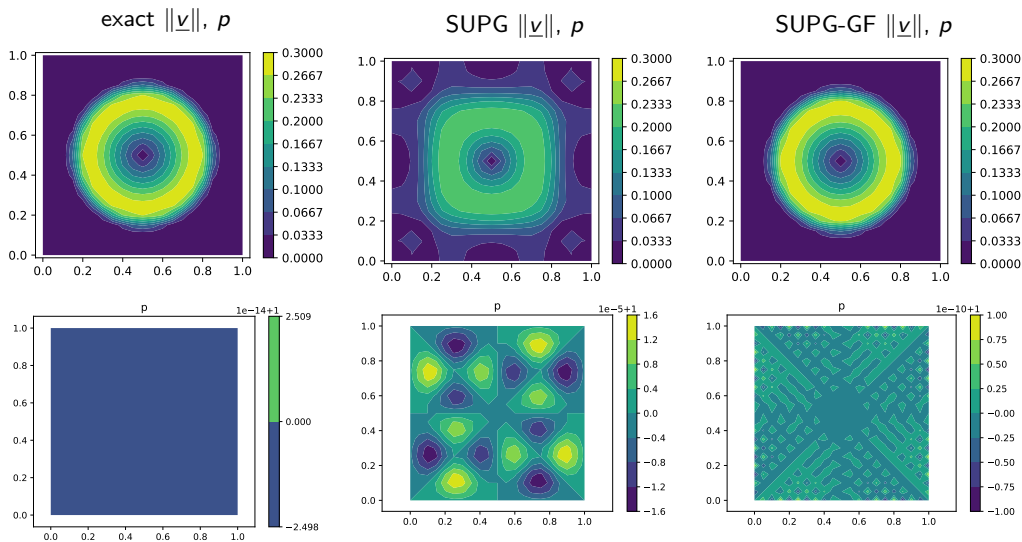
$$\partial_t \mathbf{v} + \partial_y \mathbf{p} = S_v$$

$$\partial_t \mathbf{p} + \partial_x \mathbf{u} + \partial_y \mathbf{v} = S_p$$

### Details on discretization

- Cartesian grid!!
- Gauss-Lobatto points for quadrature and Lagrange basis function
- Explicit arbitrary high order time discretization with Deferred Correction

# FEM-SUPG $\mathbb{Q}^1$ on an acoustic vortex $S = 0$



### Main idea

Define

$$\sigma_x(x, y) := \int_{y_0}^y u(x, s) ds \quad \sigma_y(x, y) := \int_{x_0}^x v(s, y) ds \quad S_p(x, y) := \int_{y_0}^y \int_{x_0}^x S_p(z, s) dz ds,$$

so that

$$\partial_t p + \partial_x u + \partial_y v = \partial_t p + \partial_{xy} \underbrace{(\sigma_x + \sigma_y - S_p)}_{=: \mathcal{F}} = 0$$

## Global flux in 2D

### Main idea

Define

$$\sigma_x(x, y) := \int_{y_0}^y u(x, s) ds \quad \sigma_y(x, y) := \int_{x_0}^x v(s, y) ds \quad S_p(x, y) := \int_{y_0}^y \int_{x_0}^x S_p(z, s) dz ds,$$

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### Global Flux SUPG for acoustics

$$\int \varphi(\partial_t u + \partial_x(p - S_u)) + \alpha \Delta x \partial_x \varphi(\partial_t p + \partial_x \partial_y \mathcal{F}) = 0$$

$$\int \varphi(\partial_t v + \partial_y(p - S_v)) + \alpha \Delta y \partial_y \varphi(\partial_t p + \partial_x \partial_y \mathcal{F}) = 0$$

$$\int \varphi(\partial_t p + \partial_x \partial_y \mathcal{F}) + \alpha \Delta x \partial_x \varphi(\partial_t u + \partial_x(p - S_u)) + \alpha \Delta y \partial_y \varphi(\partial_t v + \partial_y(p - S_v)) = 0$$

### Global Flux SUPG for acoustics

$$\sigma_x(x, y) := \int_{y_0}^y u(x, s) ds \text{ and } \sigma_y(x, y) := \int_{x_0}^x v(s, y) ds, \text{ with } \sigma_x, \sigma_y \in V_h^K(\Omega_h).$$

### Changes in equilibrium

$$\nabla \cdot \underline{v} = 0$$

$$\implies \partial_x \partial_y (\sigma_x + \sigma_y) = 0$$

$$\iff \sigma_x + \sigma_y = f(x) + g(y)$$

### Discrete equilibrium

$$\partial_x \partial_y \mathcal{F}(x_i, y_j) = 0$$

$$\implies \int_{x_0}^{x_i} \int_{y_0}^{y_j} \partial_y \partial_x \mathcal{F}(x, y) dx dy = 0 \quad \forall i, j$$

$$\implies \int_{x_0}^{x_i} \partial_x \mathcal{F}(x, y_j) dx - \int_{x_0}^{x_i} \partial_x \mathcal{F}(x, y_0) dx = 0 \quad \forall i, j$$

$$\implies \mathcal{F}(x_i, y_j) - \mathcal{F}(x_0, y_j) - \mathcal{F}(x_i, y_0) + \mathcal{F}(x_0, y_0) = 0 \quad \forall i, j$$

$$\implies \mathcal{F}(x_i, y_j) = f_i + g_j$$

### Global Flux is not global!

- In principle  $\sigma_x(x, y) = \int_{y_B}^y u(x, s) ds$  should be integrated from the beginning (bottom) of the domain  $y_B$ !
- In practice we always use  $\partial_x \partial_y \sigma_x(x, y)$  integrated in one cell!!!!
- So,

$$\sigma_x(x, y) = \int_{y_B}^y u(x, s) ds = \underbrace{\int_{y_B}^{y_0} u(x, s) ds}_{\text{constant in one cell!}} + \int_{y_0}^y u(x, s) ds$$

whatever constant we bring from outside the cell, is canceled out

$$\partial_y \sigma_x(x, y) = \partial_y \int_{y_B}^y u(x, s) ds = \partial_y \int_{y_B}^{y_0} u(x, s) ds + \partial_y \int_{y_0}^y u(x, s) ds = \partial_y \int_{y_0}^y u(x, s) ds$$

- At the discrete level we have an integral operator  $I_y$  and a differential operator  $D_y$  that together give a weird averaging operator  $D_y I_y$

## Analogy with Raviart-Thomas elements

### Global Flux velocity space

Recall that  $u, v \in \mathbb{P}^d \otimes \mathbb{P}^d$  continuous across interfaces. Then, we define

$$\begin{aligned} \int^y u(x, s) ds &\in \mathbb{P}^d \otimes \mathbb{P}^{d+1} & \sigma_x &= \Pi_{\mathbb{P}^d \otimes \mathbb{P}^d} \left( \int^y u(x, s) ds \right) \in \mathbb{P}^d \otimes \mathbb{P}^d, \\ \tilde{u} &:= \partial_y \sigma_x \in \mathbb{P}^d \otimes d\mathbb{P}^{d-1}, & \tilde{v} &:= \partial_x \sigma_y \in \mathbb{P}^{d-1} \otimes d\mathbb{P}^d. \end{aligned}$$

Now, the equation that we actually discretize is  $\partial_t p + \partial_x \tilde{u} + \partial_y \tilde{v} = 0$  for the pressure  $p$ . Here,

$$(\tilde{u}, \tilde{v}) \in \mathbb{P}^d \otimes d\mathbb{P}^{d-1} \times d\mathbb{P}^{d-1} \otimes \mathbb{P}^d = \text{RT}^d.$$

### Differences with RT

- We evolve anyway  $u, v$  in  $\mathbb{P}^d \otimes \mathbb{P}^d$  and we do not need to evolve  $\tilde{u}, \tilde{v}$  in RT space
- We use a non-standard projection (defined DoF by DoF with integration)
- The SUPG stabilization is still necessary (we run explicit methods)
- We can include sources

### Shift of paradigm for kernels

Since we have written the SUPG-GF as

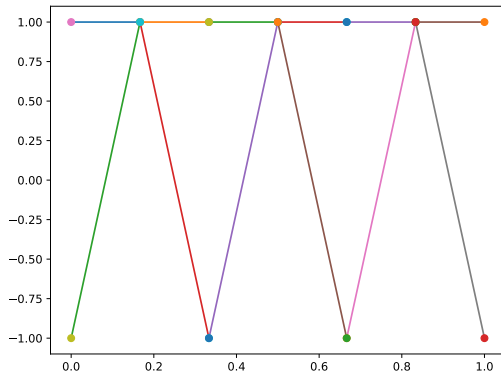
$$\int_{\Omega} (\varphi + \alpha \Delta \partial_x \varphi J^x + \alpha \Delta \partial_y \varphi J^y) (\partial_t \mathbf{q} + \partial_{xy} \mathcal{F}) = 0 \quad \forall \varphi,$$

we need to study the operators given by

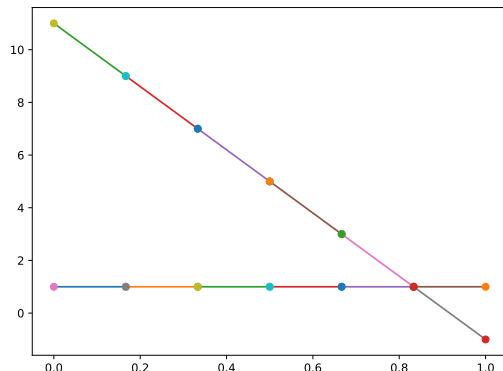
$$D_x \otimes D_y := \int_{\Omega} \varphi_i \partial_{xy} \varphi_j, \quad D_x^x \otimes D_y := \int_{\Omega} \partial_x \varphi_i \partial_{xy} \varphi_j, \quad D_x \otimes D_y^y := \int_{\Omega} \partial_y \varphi_i \partial_{xy} \varphi_j.$$

# One dimensional kernels of $D_x$ and $D_x^\times$

Kernel of  $D_x$ , 2 basis functions



Kernel of  $D_x^\times$ , 2 basis functions



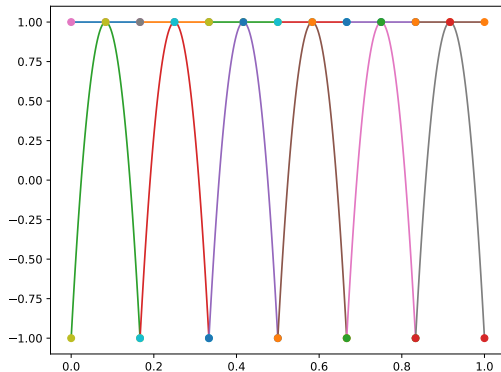
## Operators

• Divergence  $D_x \otimes D_y$

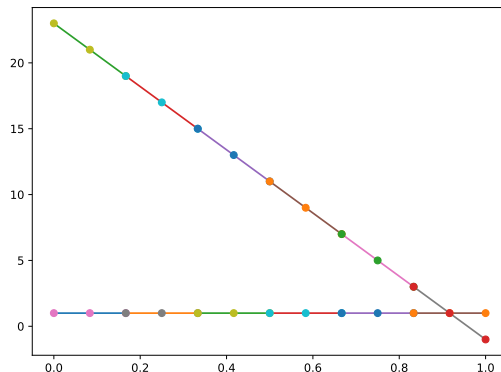
• Stabilization  $D_x^\times \otimes D_y, D_x \otimes D_y^\times$

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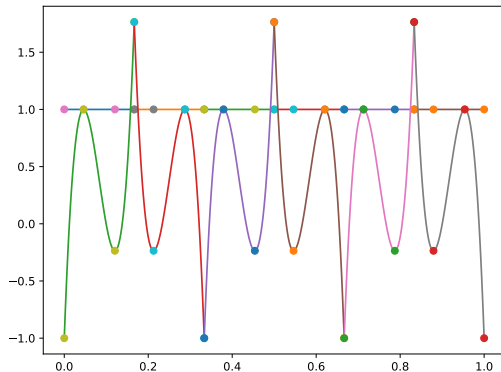
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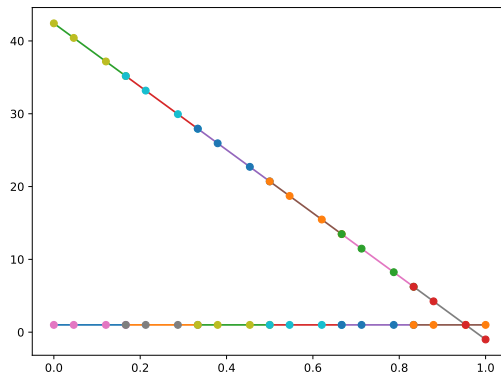
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## Operators

• Divergence  $D_x \otimes D_y$

• Stabilization  $D_x^\times \otimes D_y, D_x \otimes D_y^\times$

## Kernels of $D_x$ , $D_x^\times$ and 2D operators

---

### Theorem

- $\ker(D_x) = \text{span}\{1, w\}$  with  $w$  with first derivative discontinuous at each interface (chainsaw function)
- $\ker(D_x^\times) = \text{span}\{1, x\}$

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### Corollary

- $w \notin \ker(D_x^\times) \Rightarrow$  if stable  $D_x^\times$  should dissipate spurious oscillating modes that are in the kernel of  $D_x$

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### Kernel of $D_x \otimes D_y$

- $\ker(D_x \otimes D_y) = \ker(D_x) \otimes \mathbb{R}^{N_y K+1} + \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$

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### Kernel of $D_x \otimes D_y$

- $\ker(D_x \otimes D_y) = \ker(D_x) \otimes \mathbb{R}^{N_y K+1} + \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$
- $D_x \otimes D_y \mathcal{F} = 0 \iff \mathcal{F} = \Phi_x + \Phi_y$  with

## Kernels of $D_x$ , $D_x^x$ and 2D operators

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- $\ker(D_x^x) = \text{span}\{1, x\}$

### Corollary

- $w \notin \ker(D_x^x) \Rightarrow$  if stable  $D_x^x$  should dissipate spurious oscillating modes that are in the kernel of  $D_x$

### Kernel of $D_x \otimes D_y$

- $\ker(D_x \otimes D_y) = \ker(D_x) \otimes \mathbb{R}^{N_y K+1} + \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$
- $D_x \otimes D_y \mathcal{F} = 0 \iff \mathcal{F} = \Phi_x + \Phi_y$  with
- $\Phi_x \in \ker(D_x) \otimes \mathbb{R}^{N_y K+1}$  and
- $\Phi_y \in \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$

## Kernels of $D_x$ , $D_x^x$ and 2D operators

### Theorem

- $\ker(D_x) = \text{span}\{1, w\}$  with  $w$  with first derivative discontinuous at each interface (chainsaw function)
- $\ker(D_x^x) = \text{span}\{1, x\}$

### Corollary

- $w \notin \ker(D_x^x) \Rightarrow$  if stable  $D_x^x$  should dissipate spurious oscillating modes that are in the kernel of  $D_x$

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- $\Phi_x \in \ker(D_x) \otimes \mathbb{R}^{N_y K+1}$  and
- $\Phi_y \in \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$
- $\Phi_x = \mathbf{1} \otimes g_1 + w \otimes g_2$  with  $g_1, g_2 \in \mathbb{R}^{N_y K+1}$
- $\Phi_y = f_1 \otimes \mathbf{1} + f_2 \otimes w$  with  $f_1, f_2 \in \mathbb{R}^{N_x K+1}$

## Kernels of $D_x$ , $D_x^x$ and 2D operators

### Theorem

- $\ker(D_x) = \text{span}\{1, w\}$  with  $w$  with first derivative discontinuous at each interface (chainsaw function)
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### Kernel of $D_x \otimes D_y$

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- $\Phi_x \in \ker(D_x) \otimes \mathbb{R}^{N_y K+1}$  and
- $\Phi_y \in \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$
- $\Phi_x = 1 \otimes g_1 + w \otimes g_2$  with  $g_1, g_2 \in \mathbb{R}^{N_y K+1}$
- $\Phi_y = f_1 \otimes 1 + f_2 \otimes w$  with  $f_1, f_2 \in \mathbb{R}^{N_x K+1}$

### Corollary

- $w \notin \ker(D_x^x) \Rightarrow$  if stable  $D_x^x$  should dissipate spurious oscillating modes that are in the kernel of  $D_x$

### Two Dimensional Operators

- $\Phi_x = \underbrace{1 \otimes g_1}_{\text{Good equilibrium}} + \underbrace{w \otimes g_2}_{\text{Bad equilibrium}}$

## Kernels of $D_x$ , $D_x^\times$ and 2D operators

### Theorem

- $\ker(D_x) = \text{span}\{1, w\}$  with  $w$  with first derivative discontinuous at each interface (chainsaw function)
- $\ker(D_x^\times) = \text{span}\{1, x\}$

### Kernel of $D_x \otimes D_y$

- $\ker(D_x \otimes D_y) = \ker(D_x) \otimes \mathbb{R}^{N_y K+1} + \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$
- $D_x \otimes D_y \mathcal{F} = 0 \iff \mathcal{F} = \Phi_x + \Phi_y$  with
- $\Phi_x \in \ker(D_x) \otimes \mathbb{R}^{N_y K+1}$  and
- $\Phi_y \in \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$
- $\Phi_x = 1 \otimes g_1 + w \otimes g_2$  with  $g_1, g_2 \in \mathbb{R}^{N_y K+1}$
- $\Phi_y = f_1 \otimes 1 + f_2 \otimes w$  with  $f_1, f_2 \in \mathbb{R}^{N_x K+1}$

### Corollary

- $w \notin \ker(D_x^\times) \Rightarrow$  if stable  $D_x^\times$  should dissipate spurious oscillating modes that are in the kernel of  $D_x$

### Two Dimensional Operators

- $\Phi_x = \underbrace{1 \otimes g_1}_{\text{Good equilibrium}} + \underbrace{w \otimes g_2}_{\text{Bad equilibrium}}$
- $(D_x^\times \otimes D_y)(1 \otimes g_1) = D_x^\times 1 \otimes D_y g_1 = 0 \otimes * = 0$
- $(D_x \otimes D_y^\times)(1 \otimes g_1) = D_x 1 \otimes D_y^\times g_1 = 0 \otimes * = 0$

## Kernels of $D_x$ , $D_x^x$ and 2D operators

### Theorem

- $\ker(D_x) = \text{span}\{1, w\}$  with  $w$  with first derivative discontinuous at each interface (chainsaw function)
- $\ker(D_x^x) = \text{span}\{1, x\}$

### Kernel of $D_x \otimes D_y$

- $\ker(D_x \otimes D_y) = \ker(D_x) \otimes \mathbb{R}^{N_y K+1} + \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$
- $D_x \otimes D_y \mathcal{F} = 0 \iff \mathcal{F} = \Phi_x + \Phi_y$  with
- $\Phi_x \in \ker(D_x) \otimes \mathbb{R}^{N_y K+1}$  and
- $\Phi_y \in \mathbb{R}^{N_x K+1} \otimes \ker(D_y)$
- $\Phi_x = 1 \otimes g_1 + w \otimes g_2$  with  $g_1, g_2 \in \mathbb{R}^{N_y K+1}$
- $\Phi_y = f_1 \otimes 1 + f_2 \otimes w$  with  $f_1, f_2 \in \mathbb{R}^{N_x K+1}$

### Corollary

- $w \notin \ker(D_x^x) \Rightarrow$  if stable  $D_x^x$  should dissipate spurious oscillating modes that are in the kernel of  $D_x$

### Two Dimensional Operators

- $\Phi_x = \underbrace{1 \otimes g_1}_{\text{Good equilibrium}} + \underbrace{w \otimes g_2}_{\text{Bad equilibrium}}$
- $(D_x^x \otimes D_y)(1 \otimes g_1) = D_x^x 1 \otimes D_y g_1 = 0 \otimes * = 0$
- $(D_x \otimes D_y^y)(1 \otimes g_1) = D_x 1 \otimes D_y^y g_1 = 0 \otimes * = 0$
- $(D_x \otimes D_y^y)(w \otimes g_2) = D_x w \otimes D_y^y g_2 = 0 \otimes * = 0$
- $(D_x^x \otimes D_y)(w \otimes g_2) = D_x^x w \otimes D_y g_2 \neq 0$  except
- $(D_x^x \otimes D_y)(w \otimes 1) = D_x^x w \otimes D_y 1 = 0$  and
- $(D_x^x \otimes D_y)(w \otimes w) = D_x^x w \otimes D_y w = 0$

### GF+FEM+SUPG

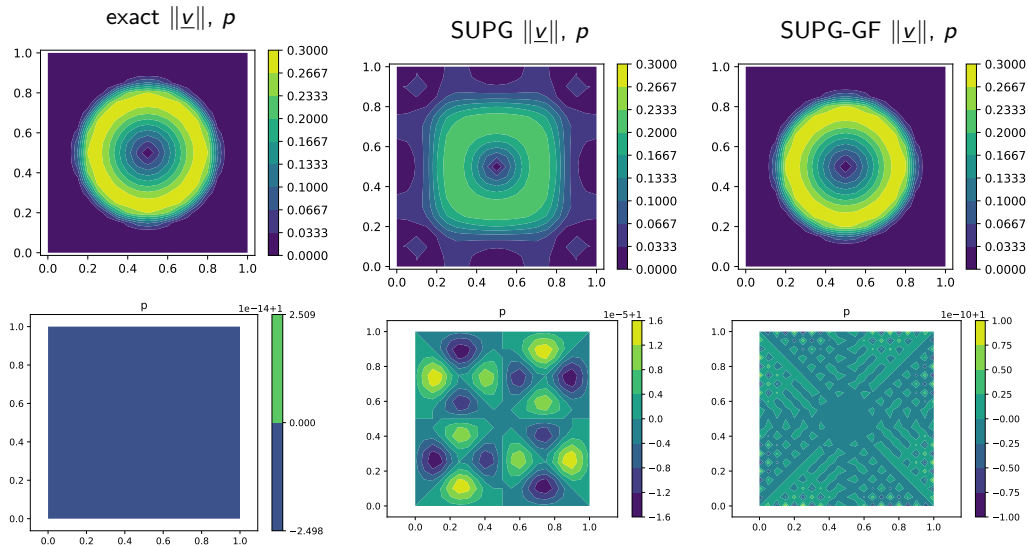
$$\partial_t \mathbf{q} + \partial_x f(\mathbf{q}) + \partial_y g(\mathbf{q}) = S(\mathbf{q}) \implies \partial_t \mathbf{q} + \partial_x \partial_y \mathcal{F} = 0$$

$$\mathcal{F} := \int^y f(\mathbf{q}) + \int^x g(\mathbf{q}) - \int^x \int^y S(\mathbf{q});$$

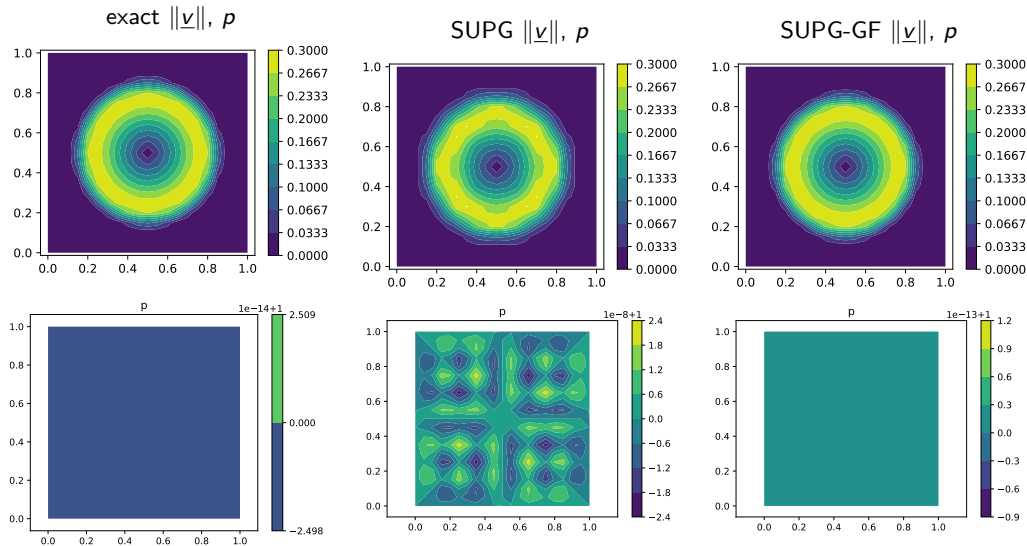
$$\int_{\Omega} (\varphi + \alpha \Delta \partial_x \varphi J^x + \alpha \Delta \partial_y \varphi J^y) (\partial_t \mathbf{q} + \partial_{xy} \mathcal{F}) = 0 \quad \forall \varphi.$$

- $J^x$  and  $J^y$  are the Jacobians of the fluxes  $f$  and  $g$ .

# Simulation of vortex (linear acoustics) FEM+SUPG: $\mathbb{Q}^1$ , $N_x = N_y = 20$



# Simulation of vortex (linear acoustics) FEM+SUPG: $\mathbb{Q}^2$ , $N_x = N_y = 10$



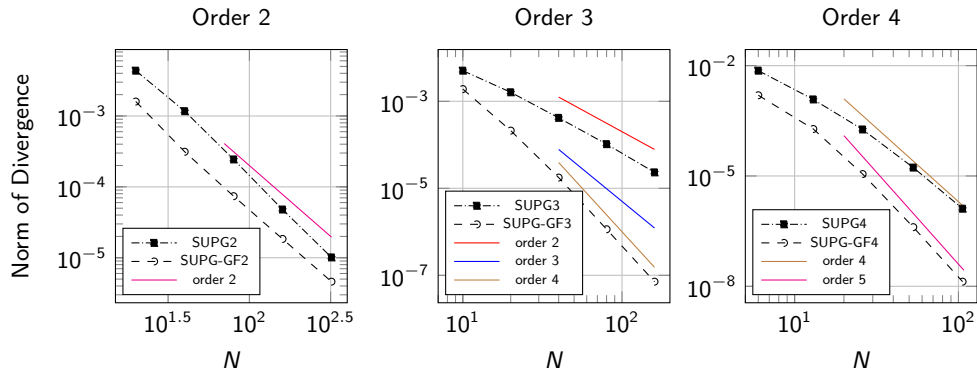


Figure: Smooth vortex: convergence of  $L^2$  error of  $u$  with respect to the number of elements in  $x$

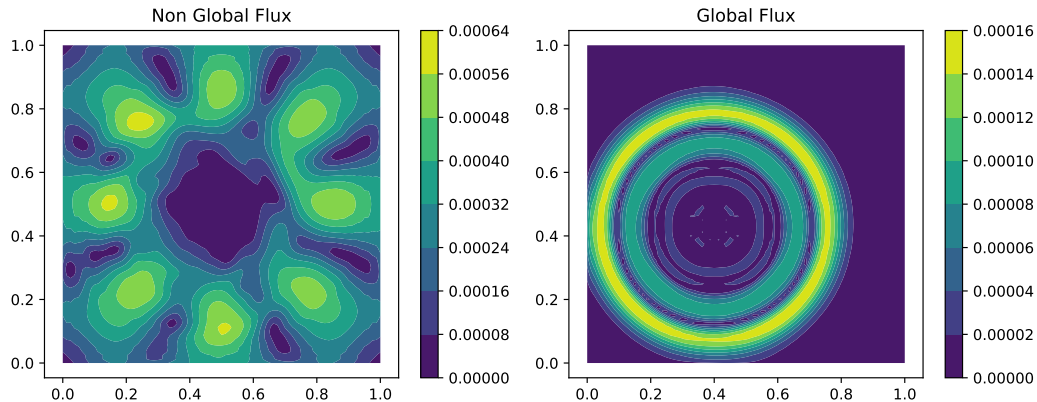
### Pressure perturbation

- Gaussian centered in  $\underline{x}_p = (0.4, 0.43)$
- scaling coefficient  $r_0 = 0.1$
- radius  $\rho(\underline{x}) = \sqrt{\|\underline{x} - \underline{x}_p\|}/r_0$

$$\delta_p(\underline{x}) = \varepsilon e^{-\frac{1}{2(1-\rho(\underline{x}))^2} + \frac{1}{2}},$$

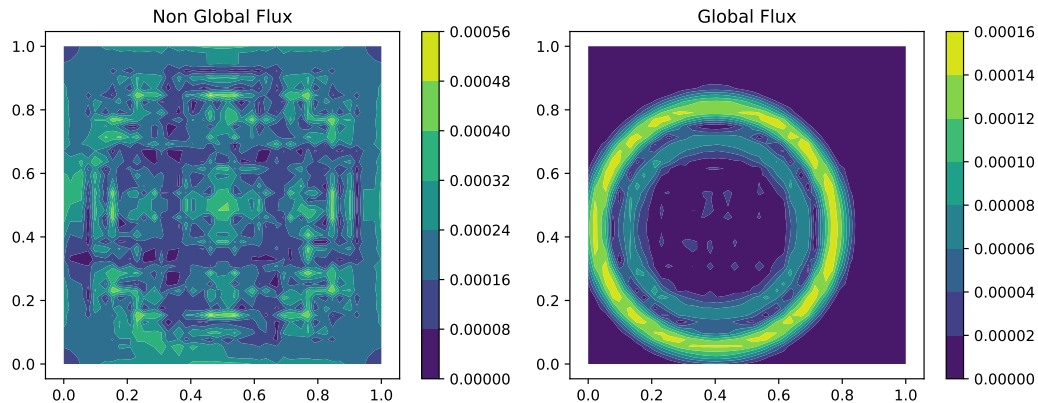
- final time  $T = 0.35$

## Vortex perturbation



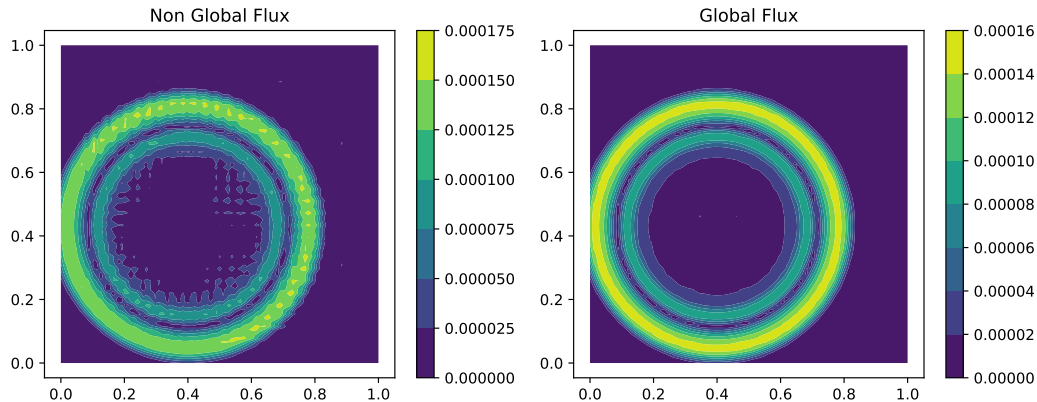
**Figure:** Perturbation( $\varepsilon = 10^{-3}$ ) test. Plot of  $\|\underline{u}_{eq} - \underline{u}_p\|$ , with  $\underline{u}_{eq}$  the equilibrium obtained with a cheap optimization process.  $\mathbb{P}^1$  with  $80 \times 80$  cells and 6561 dofs.

## Vortex perturbation



**Figure:** Perturbation( $\varepsilon = 10^{-3}$ ) test. Plot of  $\|\underline{u}_{eq} - \underline{u}_p\|$ , with  $\underline{u}_{eq}$  the equilibrium obtained with a cheap optimization process.  $\mathbb{P}^3$  with  $13 \times 13$  cells and 1600 dofs.

## Vortex perturbation



**Figure:** Perturbation( $\varepsilon = 10^{-3}$ ) test. Plot of  $\|\underline{u}_{eq} - \underline{u}_p\|$ , with  $\underline{u}_{eq}$  the equilibrium obtained with a cheap optimization process.  $\mathbb{P}^3$  with 26 cells and 6241 dofs.

### FEM-SUPG-GF

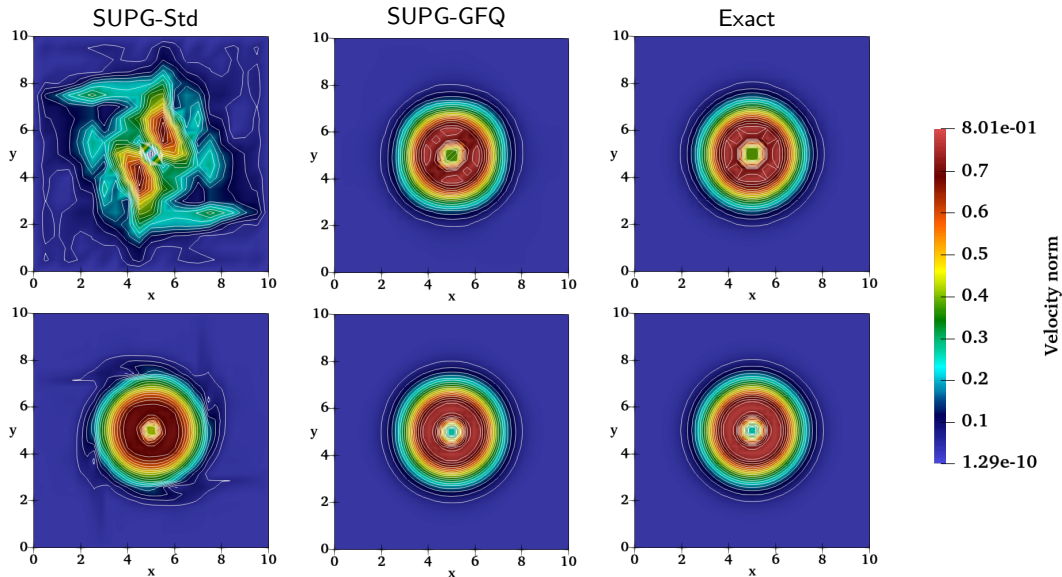
$$\partial_t u + \partial_x f(u) + \partial_y g(u) = S(u) \implies \partial_t u + \partial_x \partial_y \mathcal{G}(u) = 0$$

$$\mathcal{F}(u) := \int^y f(u) + \int^x g(u) - \int^x \int^y S(u);$$

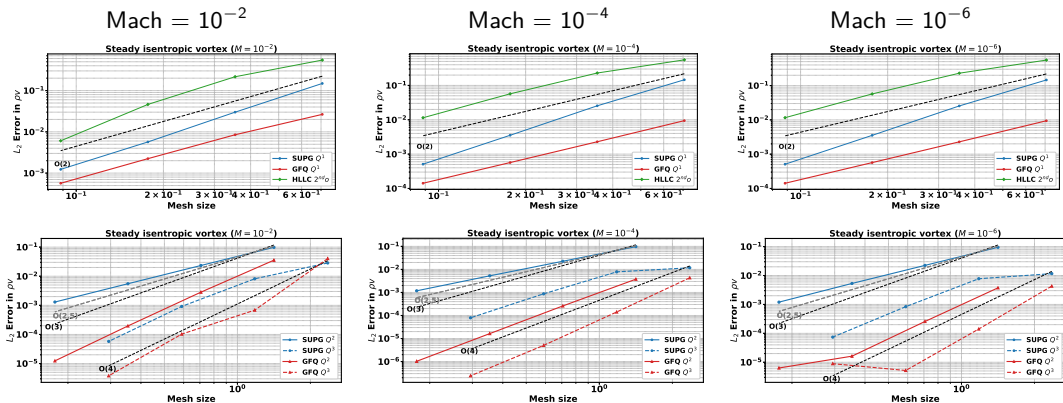
$$\int_{\Omega} (\varphi + \alpha \Delta \partial_x \varphi J^x + \alpha \Delta \partial_y \varphi J^y) (\partial_t u + \partial_{xy} \mathcal{F}(u)) = 0 \quad \forall \varphi.$$

where  $J^x$  and  $J^y$  are the Jacobians of the fluxes  $f$  and  $g$ .

Euler equations: isentropic steady vortex Mach = 0.7  $t = 50 \mathbb{Q}^1$   $N = 15, 35$  (top,bot)

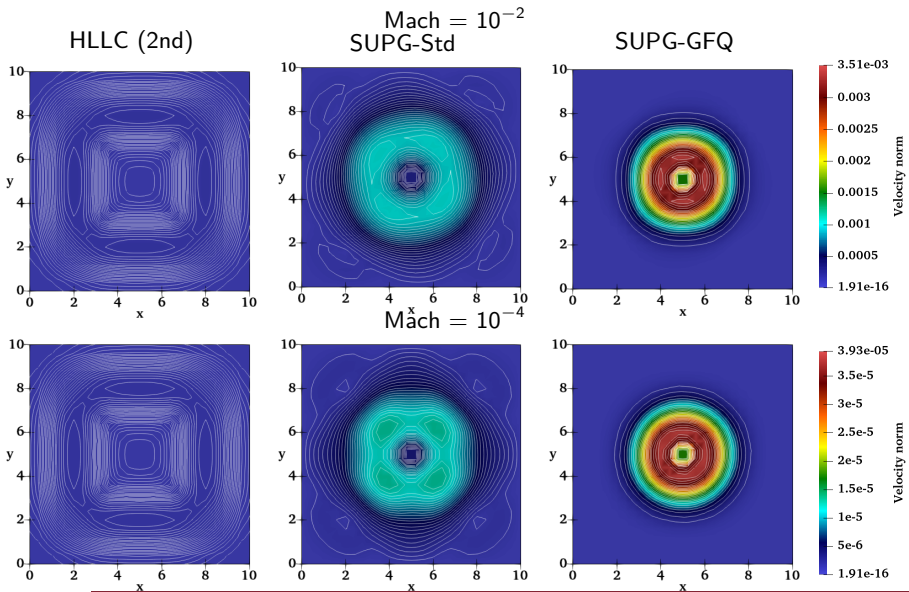


# Euler equations: isentropic steady vortex (superconvergence and low Mach)

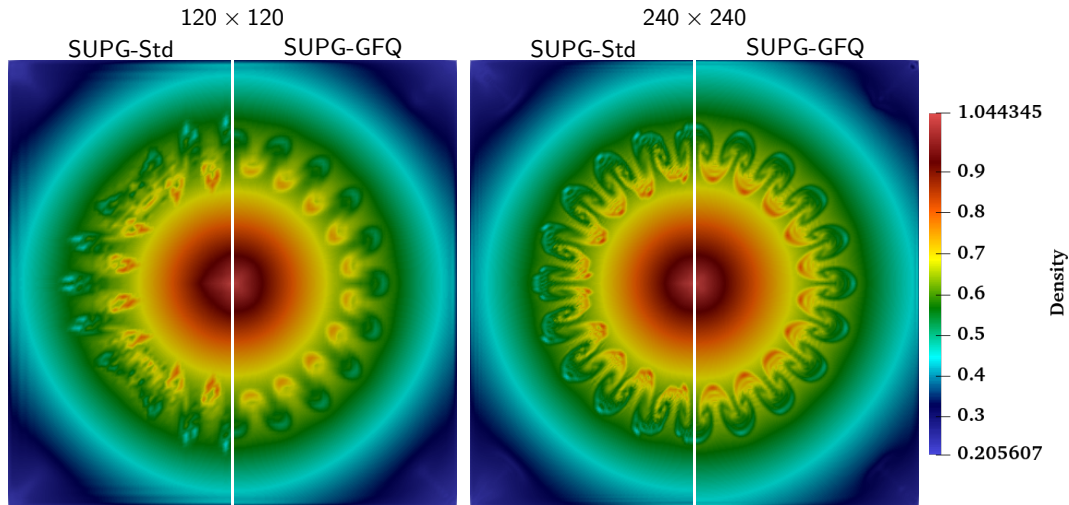


**Figure:** Convergence of the relative error on  $\rho v$  for  $Q^1$  SUPG-Std and SUPG-GFQ and for FV-HLLC (left) and  $Q^2$  and  $Q^3$  (right) at time 50 for the steady isentropic vortex test case with Mach numbers  $M = 10^{-2}$  (**top**),  $M = 10^{-4}$  (**center**) and  $M = 10^{-6}$  (**bottom**).

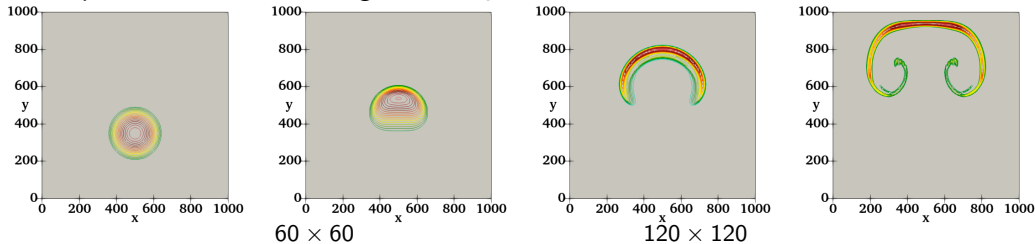
# Euler equations: isentropic steady vortex (low Mach $\mathbb{Q}^1$ $N = 25$ $t = 2000$ )



## Euler with radial gravity: Rayleigh-Taylor instability



# Euler equations: Thermal rising bubble $Q^1$ $120 \times 120$



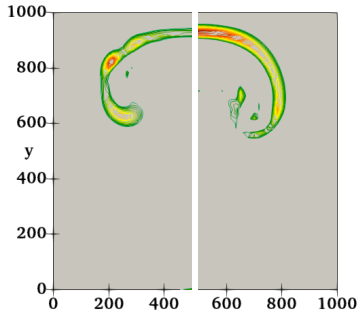
Top: GF  
150x150

SUPG-Std

SUPG-GFQ

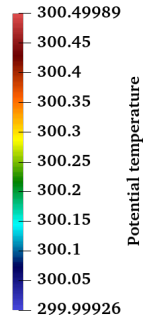
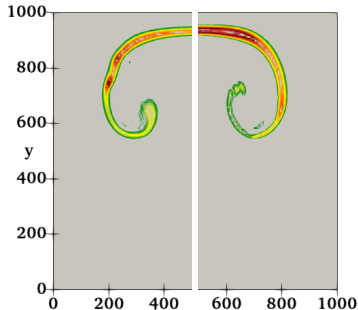
Bottom  
left: GF  
60x60

Bottom  
right:  
SUPG  
60x60



SUPG-Std

SUPG-GFQ



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- 5 Perspectives

## Finite Volume Corner-Flux Formulation

### Equation

$$\partial_t q(t, x, y) + \partial_x \partial_y \mathcal{F}(t, x, y) = 0,$$

### Finite Volume Formulation

$$\frac{q_{i,j}^{n+1} - q_{i,j}^n}{\Delta t} + \frac{\mathcal{F}(t^n, x_{i+\frac{1}{2}}, y_{j+\frac{1}{2}}) - \mathcal{F}(t^n, x_{i-\frac{1}{2}}, y_{j+\frac{1}{2}}) - \mathcal{F}(t^n, x_{i+\frac{1}{2}}, y_{j-\frac{1}{2}}) + \mathcal{F}(t^n, x_{i-\frac{1}{2}}, y_{j-\frac{1}{2}})}{\Delta x \Delta y} = 0. \quad (1)$$

$$\widehat{\mathcal{F}}_{i+\frac{1}{2}, j+\frac{1}{2}}^{(i,j)} \approx \pm \mathcal{F}(t^n, x_{i+\frac{1}{2}}, y_{j+\frac{1}{2}}) \quad (2)$$

Corner flux definition<sup>1</sup>

$$\frac{q_{i,j}^{n+1} - q_{i,j}^n}{\Delta t} + \frac{\widehat{\mathcal{F}}_{i+\frac{1}{2}, j+\frac{1}{2}}^{(i,j)} + \widehat{\mathcal{F}}_{i+\frac{1}{2}, j+\frac{1}{2}}^{(i+1,j+1)} + \widehat{\mathcal{F}}_{i+\frac{1}{2}, j+\frac{1}{2}}^{(i+1,j)} + \widehat{\mathcal{F}}_{i+\frac{1}{2}, j+\frac{1}{2}}^{(i,j+1)}}{\Delta x \Delta y} = 0. \quad (3)$$

<sup>1</sup>Barsukow, Ciallella, Riucciuto, Torlo. 2026.

## Corner Flux Sketch (Dual Cell SUPG formulation)

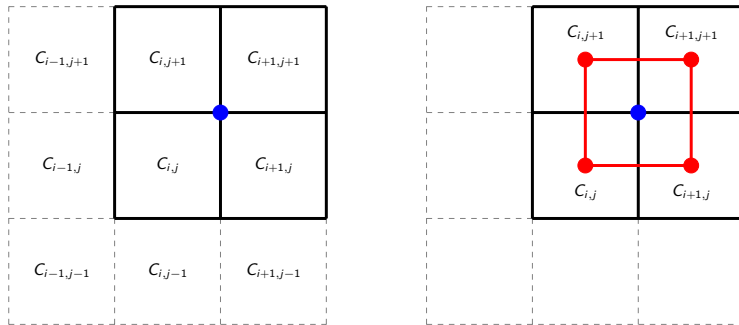


Figure: Cell labeling for the 2D grid: main grid (left), dual grid (right).

- SUPG Corner Fluxes

$$\begin{aligned}\widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)}, & \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)}, \\ \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)}, & \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)},\end{aligned}$$

- Central Part

$$\overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} = \frac{1}{4}(\mathcal{F}_{i,j} + \mathcal{F}_{i+1,j} + \mathcal{F}_{i+1,j+1} + \mathcal{F}_{i,j+1})$$

- Diffusion Part

$$\begin{aligned}\mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} &= \frac{\alpha\Delta}{4} \left( -\frac{J^x}{\Delta x} + \frac{J^y}{\Delta y} \right) \Phi_{i+\frac{1}{2},j+\frac{1}{2}}, & \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} &= \frac{\alpha\Delta}{4} \left( +\frac{J^x}{\Delta x} + \frac{J^y}{\Delta y} \right) \Phi_{i+\frac{1}{2},j+\frac{1}{2}}, \\ \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} &= \frac{\alpha\Delta}{4} \left( -\frac{J^x}{\Delta x} - \frac{J^y}{\Delta y} \right) \Phi_{i+\frac{1}{2},j+\frac{1}{2}}, & \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} &= \frac{\alpha\Delta}{4} \left( +\frac{J^x}{\Delta x} - \frac{J^y}{\Delta y} \right) \Phi_{i+\frac{1}{2},j+\frac{1}{2}},\end{aligned}$$

$$\Phi_{i+\frac{1}{2},j+\frac{1}{2}} := \Phi(\widetilde{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}) := \mathcal{F}_{i+1,j+1} - \mathcal{F}_{i,j+1} - \mathcal{F}_{i+1,j} + \mathcal{F}_{i,j} = \int_{\widetilde{\mathcal{C}}_{i+\frac{1}{2},j+\frac{1}{2}}} \partial_{xy} \widetilde{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} dx dy.$$

- When  $\mathcal{F}_{i,j} = \sigma_i + \gamma_j$  **everything cancels out!**

## Comparison with Rusanov Corner Flux

- SUPG Corner Fluxes

$$\begin{aligned}\widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)}, & \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)}, \\ \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)}, & \widehat{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} &= \overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} n_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} + \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)},\end{aligned}$$

- Central Part

$$\overline{\mathcal{F}}_{i+\frac{1}{2},j+\frac{1}{2}} = \frac{1}{4}(\mathcal{F}_{i,j} + \mathcal{F}_{i+1,j} + \mathcal{F}_{i+1,j+1} + \mathcal{F}_{i,j+1})$$

- Diffusion Part

$$\begin{aligned}\mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j+1)} &= \alpha \Delta(q_{i,j+1} - \bar{q}_{i+\frac{1}{2},j+\frac{1}{2}}), & \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j+1)} &= \alpha \Delta(q_{i+1,j+1} - \bar{q}_{i+\frac{1}{2},j+\frac{1}{2}}), \\ \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i,j)} &= \alpha \Delta(q_{i,j} - \bar{q}_{i+\frac{1}{2},j+\frac{1}{2}}), & \mathcal{D}_{i+\frac{1}{2},j+\frac{1}{2}}^{(i+1,j)} &= \alpha \Delta(q_{i+1,j} - \bar{q}_{i+\frac{1}{2},j+\frac{1}{2}}), \\ \bar{q}_{i+\frac{1}{2},j+\frac{1}{2}} &= \frac{1}{4}(q_{i,j} + q_{i+1,j} + q_{i+1,j+1} + q_{i,j+1})\end{aligned}$$

- When  $\mathcal{F}_{i,j} = \sigma_i + \gamma_j$  **the diffusion part might still contribute!**

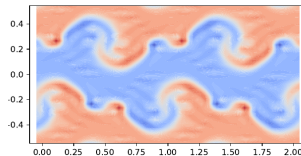
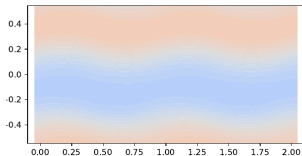
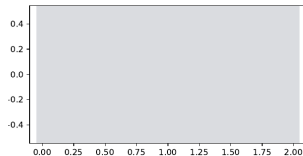
## Euler equations: Kevin-Helmoltz instability

$N_x = 64, N_y = 32$

FV-1

FV-2

GF

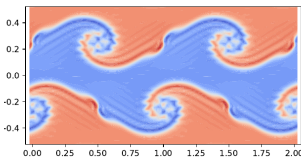
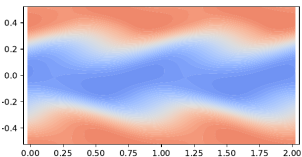
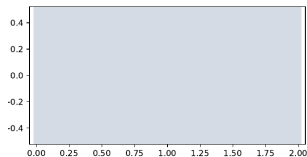


$N_x = 128, N_y = 64$

FV-1

FV-2

GF



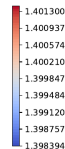
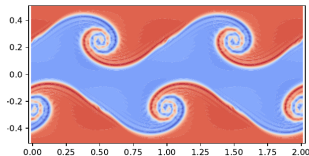
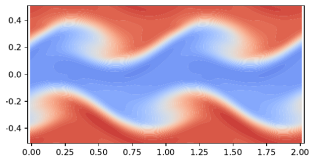
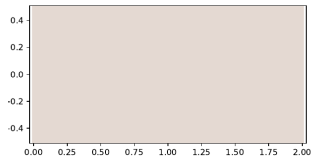
## Euler equations: Kevin-Helmoltz instability

$$N_x = 256, N_y = 128$$

FV-1

FV-2

GF

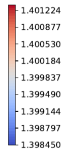
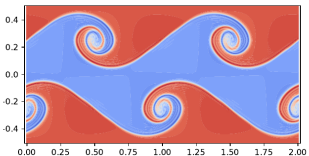
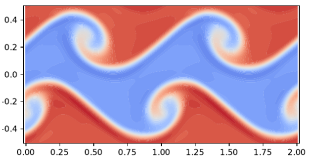
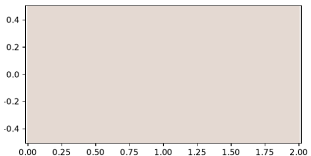


$$N_x = 512, N_y = 256$$

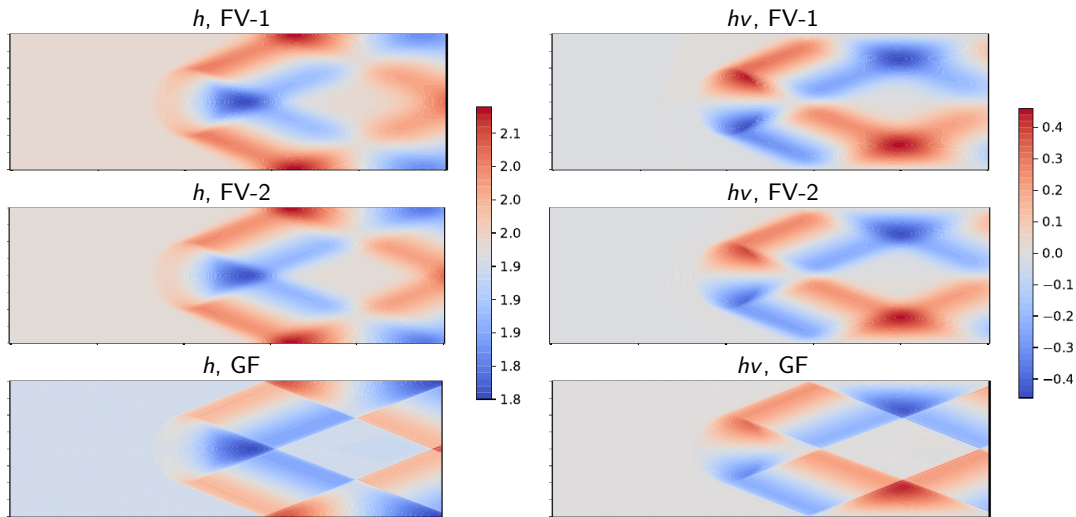
FV-1

FV-2

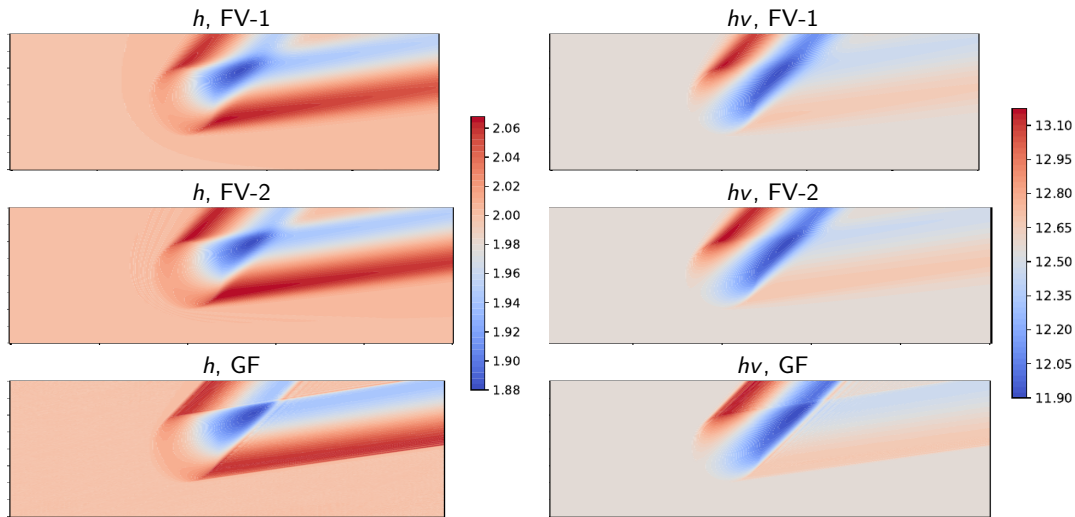
GF



## Shallow water: subcritical flow with bathymetry



## Shallow water: subcritical flow with bathymetry

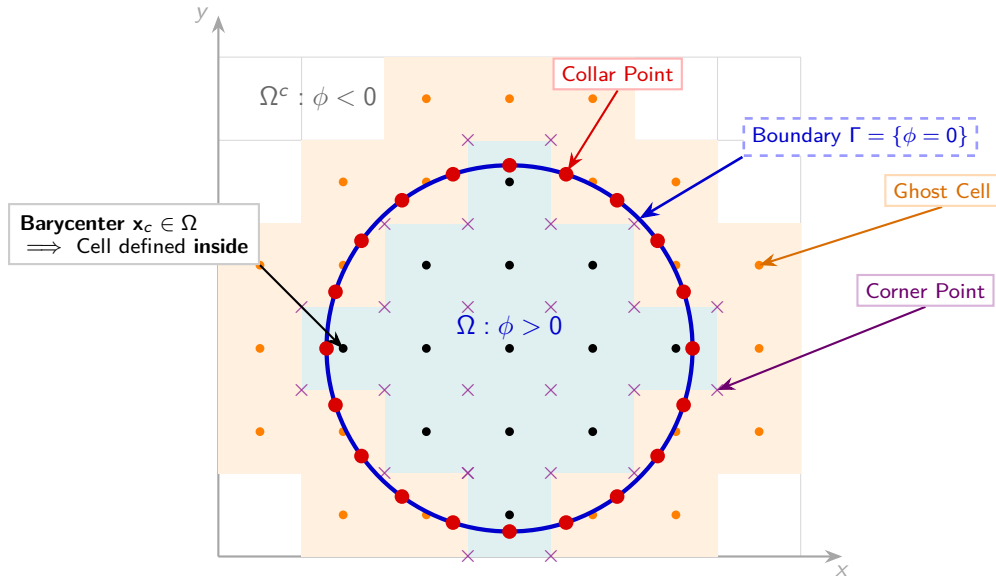


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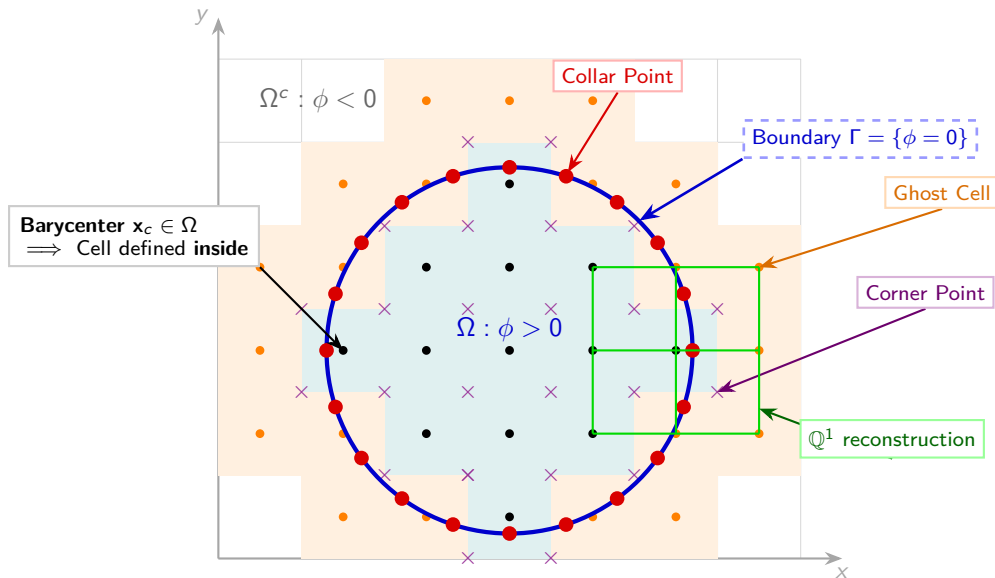
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## Geometric Classifications: How to set ghosts cells?



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## Extrapolation from the Interior and Collar Points

### Least Square Problem (Dirichlet BC)

$$\min_{p(x,y) \in \mathbb{Q}^1} \sum_{\ell=-1}^1 \sum_{r=-1}^1 (p(x_{i+\ell}^r, y_{j+r}^r) - q(x_{i+\ell}^r, y_{j+r}^r)) \chi(x_{i+\ell}^r, y_{j+r}^r)_{\tilde{\Omega}} \quad \text{subject to} \quad (4)$$
$$p(x_k^c, y_k^c) = u(x_k^c, y_k^c), \quad (x_k^c, y_k^c) \in \Gamma \text{ collar points for } k = 1, \dots, N_{i+\frac{1}{2}, j+\frac{1}{2}}^c,$$

where  $(x_k^c, y_k^c)$  are the closest  $N_{i+\frac{1}{2}, j+\frac{1}{2}}^c \in \mathbb{N}$  collar points to the corner point  $(x_{i+\frac{1}{2}}, y_{j+\frac{1}{2}})$  and  $\chi(x_{i+\ell}^r, y_{j+r}^r)_{\tilde{\Omega}}$  is the indicator function that is 1 if the cell with barycenter  $(x_{i+\ell}^r, y_{j+r}^r)$  is in the computational domain and 0 otherwise.

### Computational Aspects

- Lagrange Multiplier Method
- Cheap
- Least Square Matrices are small
- **Second Order Accurate**
- Can be computed at the beginning of the simulation (and inverted)
- Number of collar points chosen to stabilize and make solvable the system

### Neumann BC

$$\min_{p(x,y) \in \mathbb{Q}^1} \sum_{\ell=-1}^1 \sum_{r=-1}^1 \left( p(x_{i+\ell}^c, y_{j+r}^c) - q(x_{i+\ell}^c, y_{j+r}^c) \right) \chi(x_{i+\ell}^c, y_{j+r}^c)_{\tilde{\Omega}} \quad \text{subject to} \quad (5)$$

$$\nabla p(x_k^c, y_k^c) \cdot \mathbf{n}(x_k^c, y_k^c) = g(x_k^c, y_k^c), \quad (x_k^c, y_k^c) \in \Gamma \text{ collar points for } k = 1, \dots, N_{i+\frac{1}{2}, j+\frac{1}{2}}^c,$$

where  $g$  is the function that defines the Neumann boundary condition, i.e.,  $\nabla u \cdot \mathbf{n} = g$  on  $\Gamma$ .

- **First Order Accurate**

### Wall BC Collar Points Constraints

- $\mathbf{v} \cdot \mathbf{n} = 0$
- $\partial_{\mathbf{n}} p = 0$
- $\partial_{\mathbf{n}}(\mathbf{v} \cdot \boldsymbol{\tau}) = 0$
- Larger system, similar blocks
- **First Order Accurate**

## Exact Solution

$$\Omega = \{1 \leq \sqrt{x^2 + y^2} \leq 1.384\},$$

$$\rho(x, y) = 1,$$

$$u(x, y) = yf(\rho(x, y)),$$

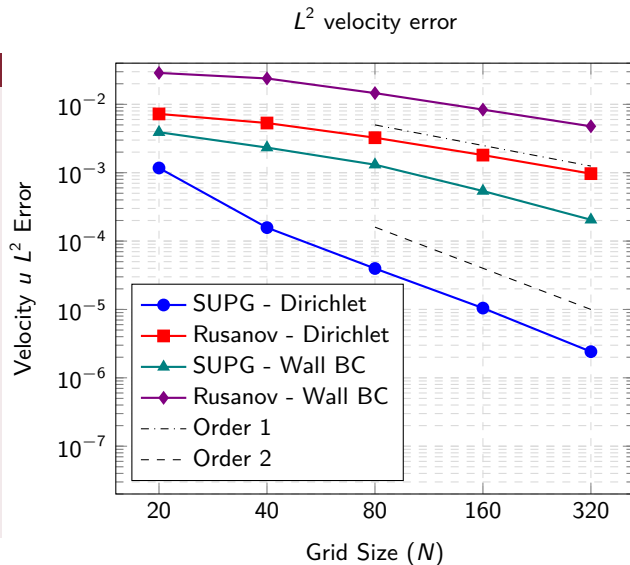
$$v(x, y) = -xf(\rho(x, y)),$$

$$r_0 = 2.5,$$

$$\rho(x, y) = \frac{\sqrt{x^2 + y^2}}{r_0},$$

$$f(\rho) = \gamma(1 + \cos(\pi\rho))^2,$$

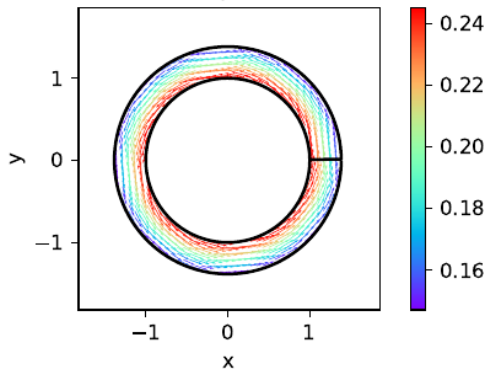
$$\gamma = \frac{12\pi\sqrt{0.981}}{r_0\sqrt{315\pi^2 - 2048}}.$$



## Acoustic vortex in ring

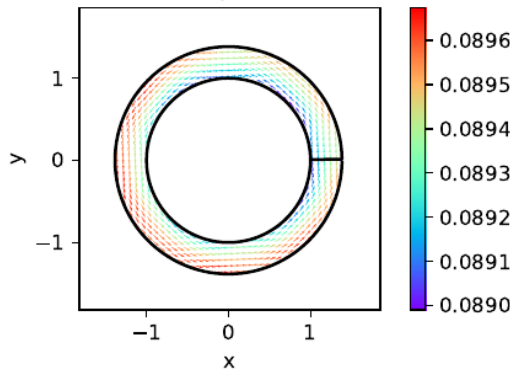
GF-SUPG-FV - GPM - Wall BC,  $N_x = N_y = 40$

Velocity vector



GF-Rusanov-FV - GPM - Wall BC,  $N_x = N_y = 40$

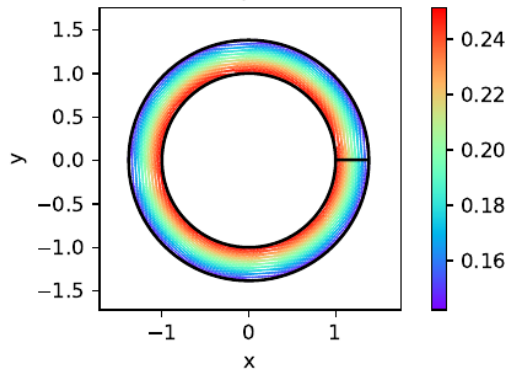
Velocity vector



## Acoustic vortex in ring

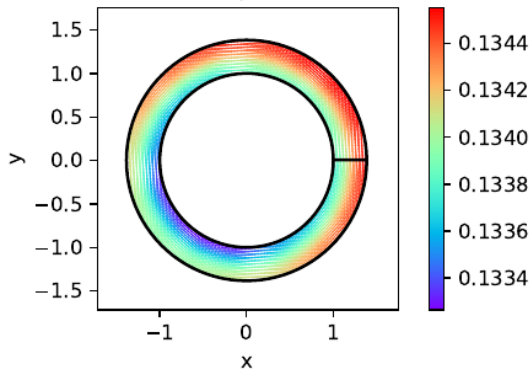
GF-SUPG-FV - GPM - Wall BC,  $N_x = N_y = 80$

Velocity vector



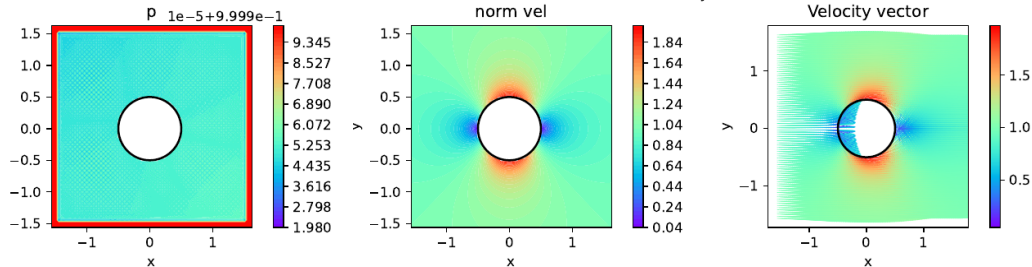
GF-Rusanov-FV - GPM - Wall BC,  $N_x = N_y = 80$

Velocity vector

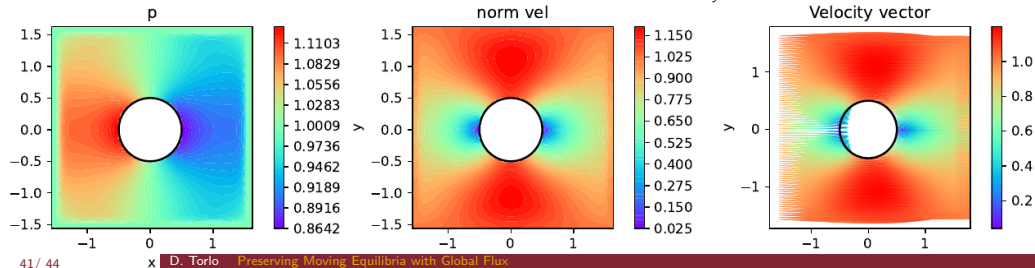


# Potential Flow Around Cylinder

GF-SUPG-FV - GPM - Wall BC,  $N_x = N_y = 80$

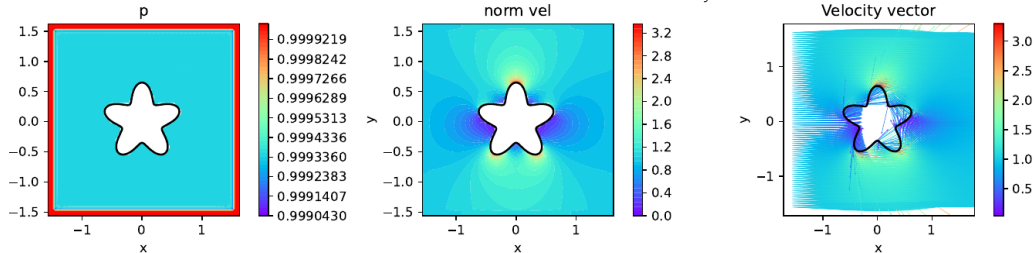


GF-Rusanov-FV - GPM - Wall BC,  $N_x = N_y = 80$

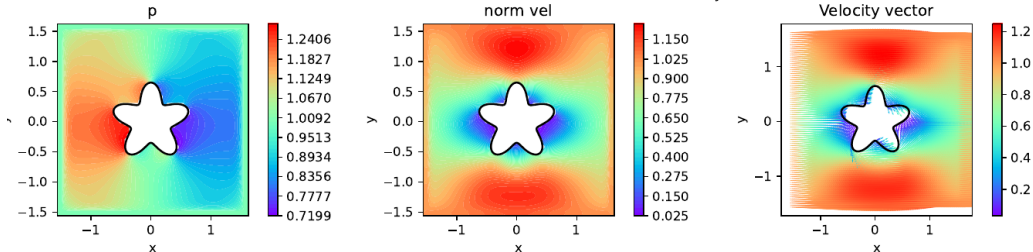


# Potential Flow Around Flower

GF-SUPG-FV - GPM - Wall BC,  $N_x = N_y = 80$



GF-Rusanov-FV - GPM - Wall BC,  $N_x = N_y = 80$



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## Extensions and Perspectives

### Summary

- Global Flux to preserve moving equilibria
- 1D integrate the source and unique flux
- 2D integrate  $f$  in  $y$  and  $g$  in  $x$
- Some superconvergence in steady states
- Extra accuracy in vorticity like problems
- Extra accuracy in low Mach problems
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- Shifted Boundary methods for FEM
- Other Riemann solver for corner problems
- Non Cartesian meshes

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THANKS!!